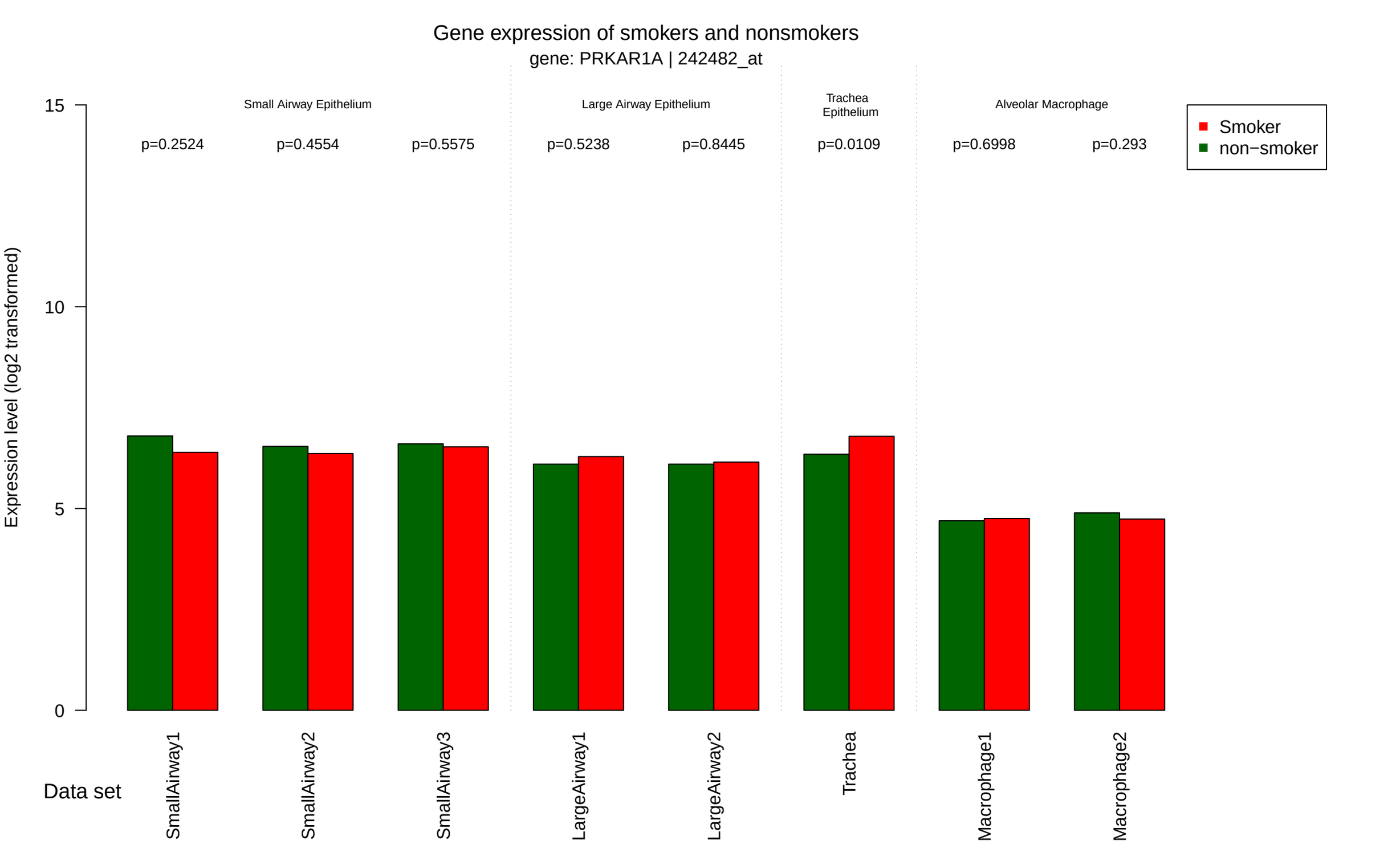
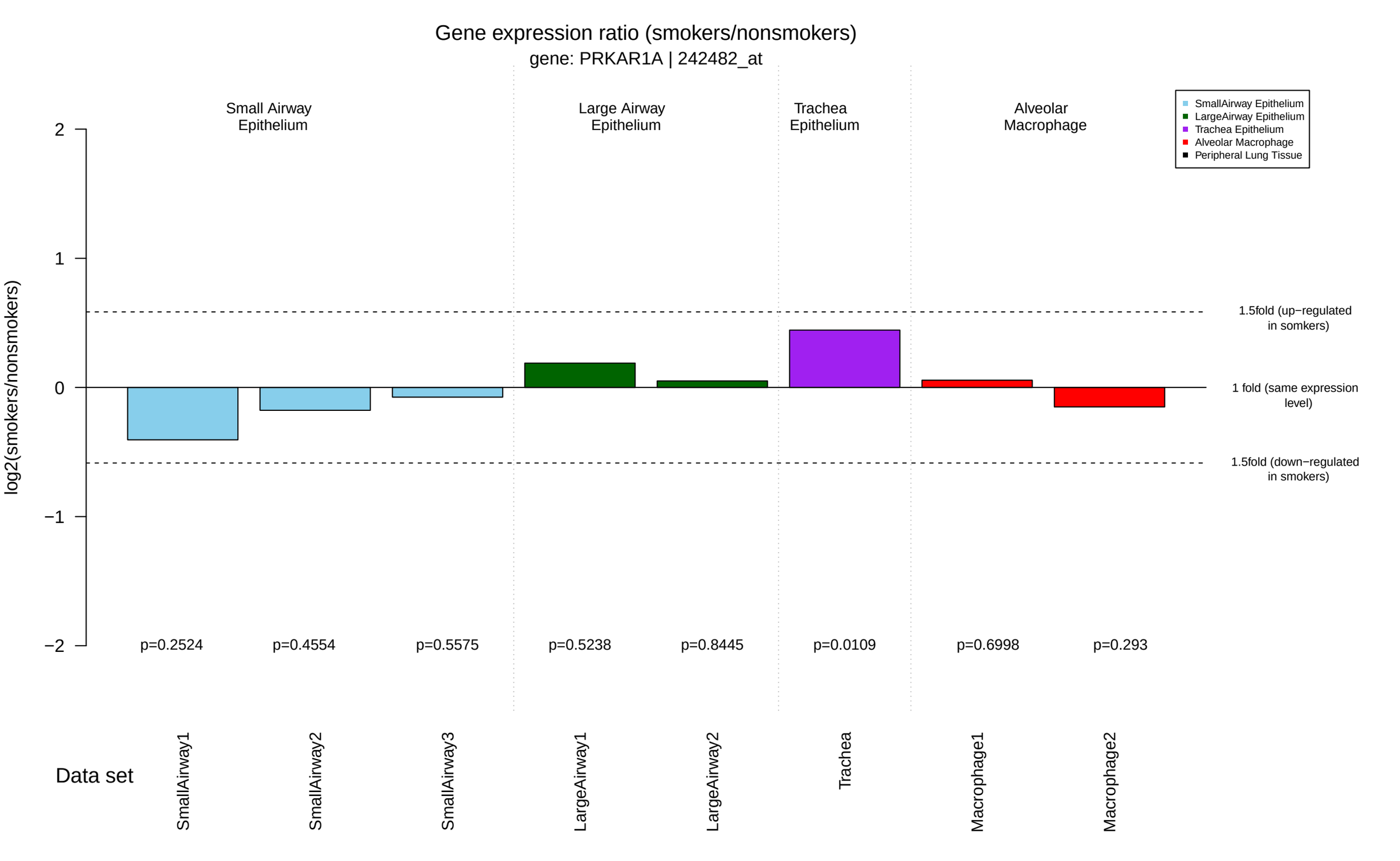
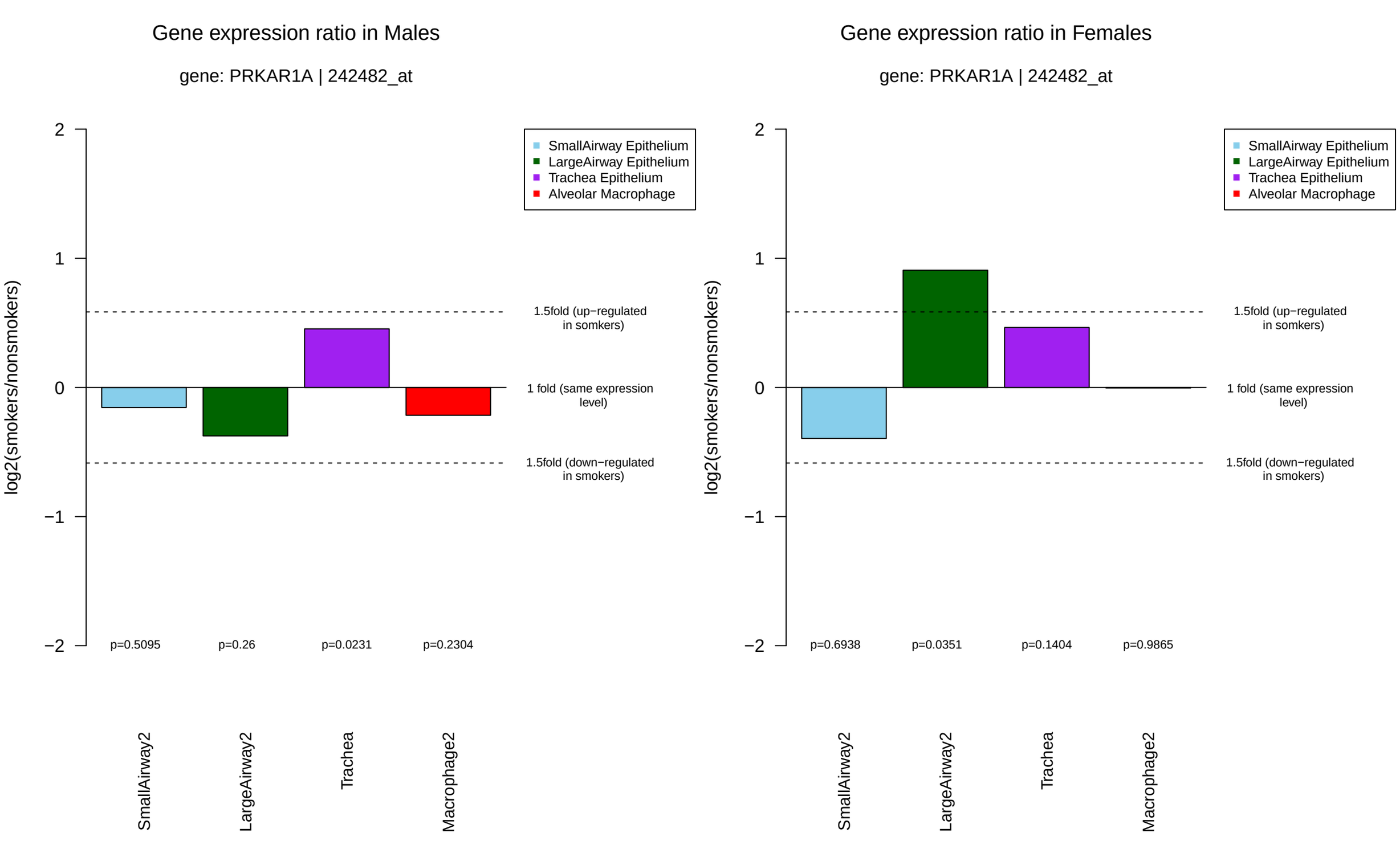
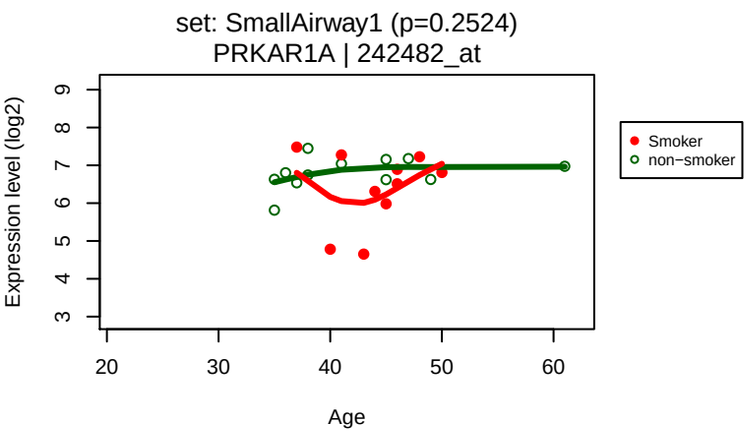
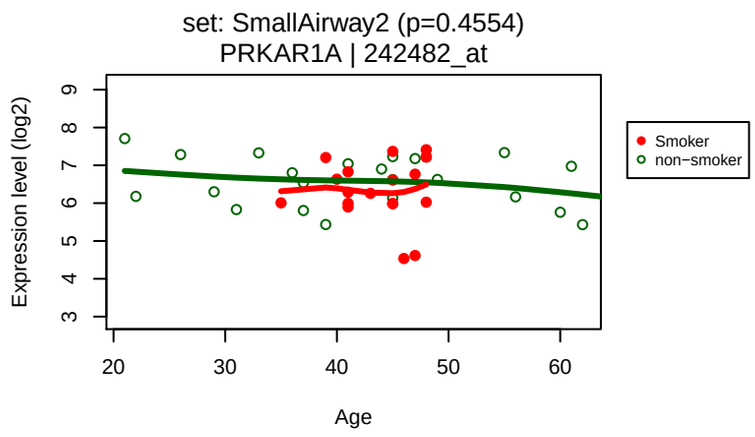
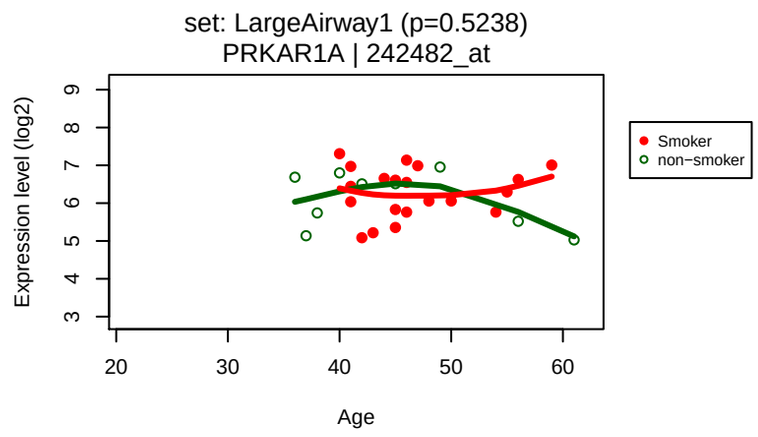

Gene expression of smokers and nonsmokers
gene: PRKAR1A | 242482_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2524
p=0.4554
p=0.5575
p=0.5238
p=0.8445
p=0.0109
p=0.6998
p=0.293
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

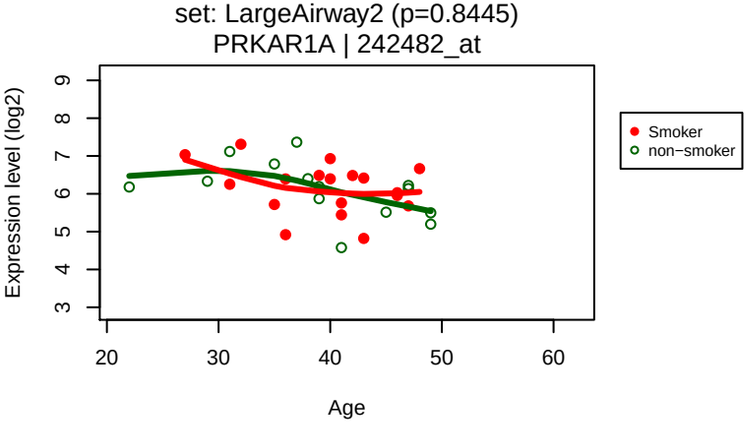

Gene expression ratio (smokers/nonsmokers)
gene: PRKAR1A | 242482_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2524
p=0.4554
p=0.5575
p=0.5238
p=0.8445
p=0.0109
p=0.6998
p=0.293
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

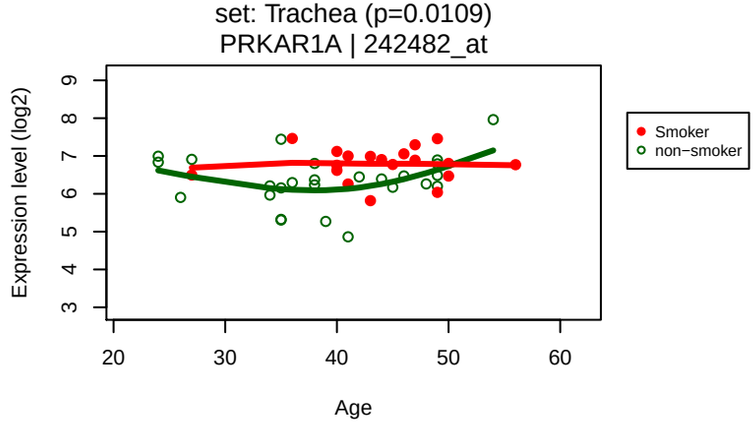

Gene expression ratio in Males
gene: PRKAR1A | 242482_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5095
p=0.26
p=0.0231
p=0.2304
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAR1A | 242482_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6938
p=0.0351
p=0.1404
p=0.9865
SmallAirway2
LargeAirway2
Trachea
Macrophage2

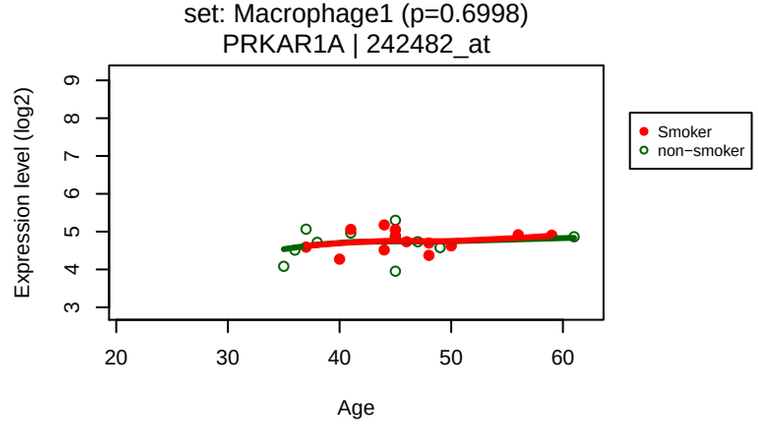

set: SmallAirway1 (p=0.2524)
PRKAR1A | 242482_at
Expression level (log2)
Age
Smoker
non−smoker

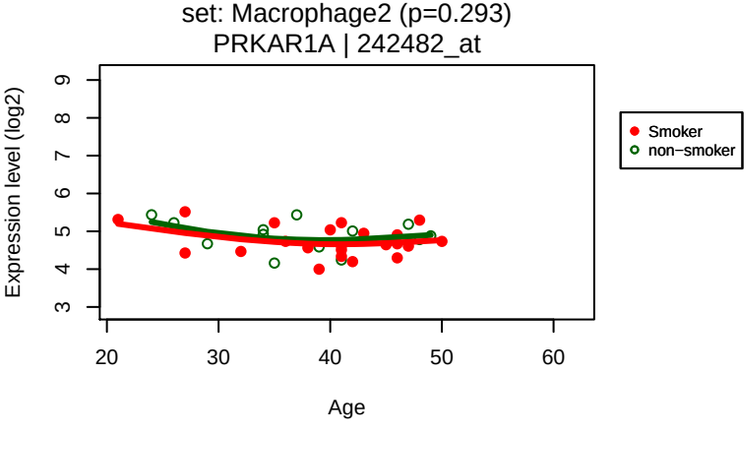

set: SmallAirway2 (p=0.4554)
PRKAR1A | 242482_at
Expression level (log2)
Age
Smoker
non−smoker

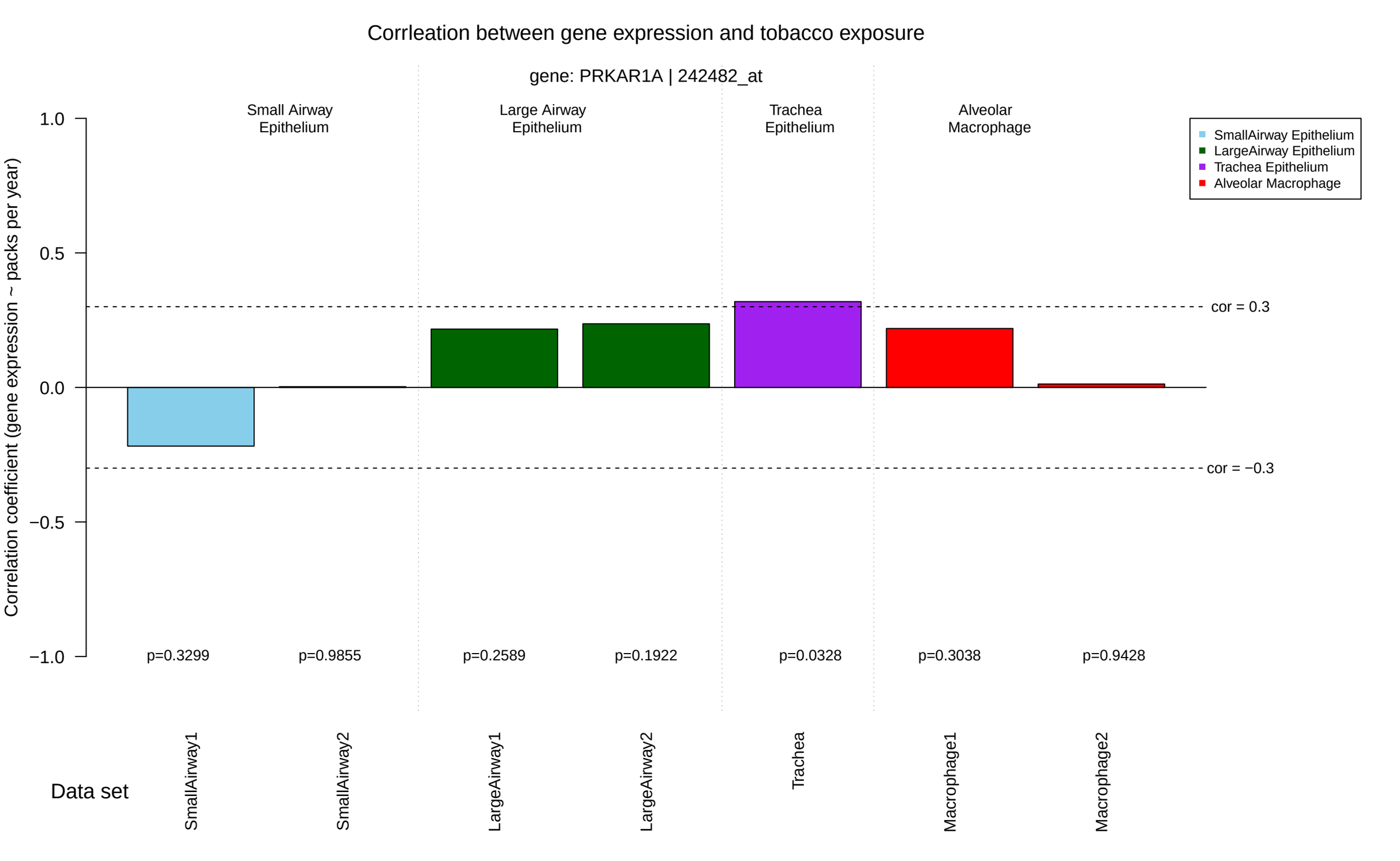

set: LargeAirway1 (p=0.5238)
PRKAR1A | 242482_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8445)
PRKAR1A | 242482_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0109)
PRKAR1A | 242482_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6998)
PRKAR1A | 242482_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.293)
PRKAR1A | 242482_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAR1A | 242482_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3299
p=0.9855
p=0.2589
p=0.1922
p=0.0328
p=0.3038
p=0.9428
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2